Day 3: Introduction to IP Addressing Theory (1 hou:IPv4 and IPv6: Differences, structure, and importance. Subnetting basics: CIDR, subnet masks. Practical (1 hour): Assign static IPs to devices and verify connectivity. Task: Solve subnetting exercises and calculate valid subnets.

Tasks

1.List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

ANS:

IPv4 vs IPv6 Comparison Table

IPv4	IPv6
1. IPv4 uses 32-bit addresses.	1. IPv6 uses 128-bit addresses.
2. It has four numbers separated by dots.	2. It uses letters and numbers separated by colons.
3. Example: 192.168.1.1	3. Example: 2001:db8::1
4. It provides fewer IP addresses.	4. It provides a very large number of IP addresses.
5. It supports broadcast.	5. It uses multicast instead of broadcast.
6. NAT is commonly used.	6. NAT is generally not required.

Real-Life Example

- **IPv4:** My home Wi-Fi router may have an address like 192.168.1.1.
- **IPv6:** Modern devices can use IPv6 addresses to communicate over the internet.

2.Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

ANS:

IP Address Assignment Table

Network: 192.168.10.0/24

Device	IP Address	Subnet Mask	Default Gateway

Laptop	192.168.10.2	255.255.255.0	192.168.10.1
Phone	192.168.10.3	255.255.255.0	192.168.10.1
Smart TV	192.168.10.4	255.255.255.0	192.168.10.1
Printer	192.168.10.5	255.255.255.0	192.168.10.1

Explanation in Simple Words

1. **Laptop:** Gets IP address 192.168.10.2.
2. **Phone:** Gets IP address 192.168.10.3.
3. **Smart TV:** Gets IP address 192.168.10.4.
4. **Printer:** Gets IP address 192.168.10.5.

Subnet Mask: 255.255.255.0 – It identifies the network part of the IP address.

Default Gateway: 192.168.10.1 – It is the router address that connects devices to the internet.

3.Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

ANS:

1. Basic Calculation

- Network: 192.168.1.0/24
- CIDR: /27
- Subnet Mask: 255.255.255.224
- Total IP addresses per subnet: 32
- Usable host addresses: 30

Number of Subnets:

\[2^{(27-24)} = 2^3 = 8 \]

Usable Hosts per Subnet:

\[2^{(32-27)}-2 = 2^5-2 = 30 \]

8 Subnets

30 Usable Hosts Each

2. First Three Subnet Ranges

Subnet	Network Address	Usable Host IPs	Broadcast Address
1	192.168.1.0	192.168.1.1 – 192.168.1.30	192.168.1.31
2	192.168.1.32	192.168.1.33 – 192.168.1.62	192.168.1.63
3	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95

3. Simple Explanation

1. /27 means 27 bits are used for the network.
2. Each subnet contains 32 IP addresses.
3. Out of 32 addresses, 2 are reserved (network and broadcast).
4. Therefore, 30 IP addresses can be used by devices.
5. The subnet address increases by 32 each time.

4.You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Hint: Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.

ANS:

Given Information

- IP Address: 10.0.5.17
- Subnet Mask: 255.255.255.0
- CIDR Notation: /24

Answer

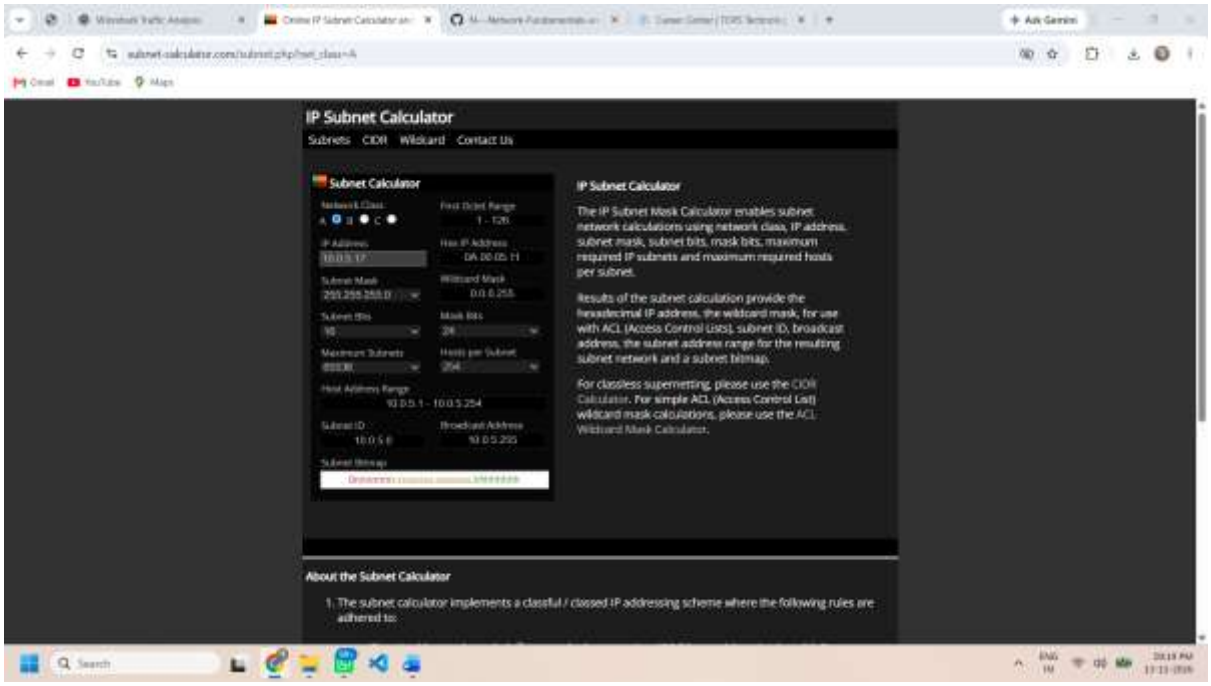
Details	IP Address
Network Address	10.0.5.0
Broadcast Address	10.0.5.255
First Valid Host	10.0.5.1
Last Valid Host	10.0.5.254
Total Usable Hosts	254

Simple Explanation

1. **Network Address:** 10.0.5.0 – Identifies the network.
2. **Broadcast Address:** 10.0.5.255 – Sends data to all devices in the network.
3. **Valid Host Range:** 10.0.5.1 to 10.0.5.254 – IP addresses available for devices.
4. **CIDR:** /24 means the first 24 bits are used for the network.

5.Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

ANS:



1. Open the website: https://www.subnet-calculator.com/
2. Enter IP Address: 10.0.5.17
3. Enter Subnet Mask: 255.255.255.0
4. Click Calculate.
5. Check the network address, broadcast address, and host range.